Data Extraction in ETL – Detailed Explanations

This document provides in-depth explanations of data extraction concepts used in ETL processes. It is suitable for academic assignments, PwSkills coursework, and interview preparation.

Question 1: Types of Data Sources in ETL

ETL systems work with multiple data sources. Common data sources include relational databases, flat files such as CSV and Excel, APIs for web-based data, cloud storage platforms, and streaming systems. Relational databases store structured data in tables. Flat files are simple and portable. APIs enable automated and real-time data access, while cloud and streaming sources support scalability.

Question 2: Data Extraction and Its Role in ETL

Data extraction is the first phase of the ETL process. It involves collecting raw data from different source systems. A successful extraction ensures data accuracy, completeness, and consistency. Errors during extraction propagate to later stages, making this step extremely critical.

Question 3: CSV vs Excel in ETL

CSV files are text-based and lightweight, making them highly suitable for ETL pipelines. Excel files support formatting and multiple sheets but consume more memory. For large-scale data processing, CSV files are preferred due to better performance and automation support.

Question 4: Extracting Data from Relational Databases

The extraction process begins with identifying the source database and establishing a secure connection. ETL developers write SQL queries to fetch required data. Extracted data is validated and stored temporarily in a staging area before transformation.

Question 5: Challenges in Data Extraction

Data quality issues such as missing or duplicate records are common challenges. Performance problems occur when dealing with large datasets. Security and compliance issues must also be handled carefully to protect sensitive data.

Question 6: APIs and Real-Time Data Extraction

APIs allow systems to exchange data in real time. They return data in formats like JSON or XML. APIs are widely used in dashboards, mobile apps, and live monitoring systems.

Question 7: Enterprise Preference for Databases

Databases are preferred in enterprise environments due to strong security, scalability, data integrity, and optimized querying capabilities. They support concurrent users efficiently.

Question 8: Handling Large CSV Files in ETL

When working with large CSV files, ETL developers should process data in chunks, use streaming techniques, optimize memory usage, log errors, and apply parallel processing. These practices ensure stable and efficient extraction.